Results in Engineering 21 (2024) 101892

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering

Full Length Article

3DUV-NetR+: A 3D hybrid semantic architecture using transformers for brain tumor segmentation with MultiModal MR images

Ilyasse Aboussaleh *, Jamal Riffi, Khalid el Fazazy, Adnane Mohamed Mahraz, Hamid Tairi

LISAC Laboratory, Department of Computer Sciences, Faculty of Sciences, Dhar El Mahraz, Sidi Mohamed ben Abdellah University, Fez, 30050, Morocco

ARTICLE INFO

Keywords:
Semantic segmentation
Deep learning
U-net
V-net
Attention mechanism
Transformers

ABSTRACT

Brain tumor segmentation plays a substantial role in Medical Image Analysis (MIS). In this regard, automatic segmentation methods facilitate precise and efficient segmentation, significantly contributing to diagnosis and treatment planning in medical applications. Recently, several Deep Learning-based architectures have been proposed to revolutionize the MIS field. Particularly, the combination of Convolution Neural Networks (CNNs) and Transformers has greatly enhanced and developed segmentation results. Moreover, the Attention mechanism in Transformers allows the modeling of long-range contextual features extracted from CNNs' encoder part. This paper proposes a hybrid advanced 3D model for brain tumor segmentation using multi-modal magnetic resonance images. The model benefits from the features extracted from the encoder of 3DU-Net and V-Net architectures at each depth. Then, a concatenation between these features and their fusion is carried out at each decoder depth to build new significant features followed by a 3D convolution layer and Transformers block for more contextual information. In addition, a final convolution block is applied to get the segmented tumor. To this end, the model is evaluated on the BraTS 2020 dataset to segment different sub-regions of brain tumors. The obtained results demonstrate the effectiveness of the proposed model in terms of dice similarity coefficient (DSC) and Hausdorff Distance (HD). For DSC, 91.95% and 82.80% and 81.70% for Whole Tumor(WT), Tumor Core (TC), and Enhancing Tumor(ET), respectively are archived, while for HD, 4.9 mm, 6.0 mm and 3.8 mm for WT, TC and ET are accomplished.

1. Introduction

Medical image analysis exploits advanced computer technology to extract, interpret, and derive meaningful information from medical images. These images, obtained through techniques such as X-rays, computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, and positron emission tomography (PET), are critical for the diagnosis, monitoring, and treatment planning of a variety of medical conditions. The field of MIS includes several key tasks such as image segmentation [1–3], classification and diagnosis [4,5], registration [6,7], image reconstruction, and image preprocessing... Notably, brain tumor segmentation presents a challenging aspect in the MIS field due to the diverse characteristics encompassing varying shapes, sizes, and appearances of tumors among individual patients. This intricate task involves identifying and delineating specific regions within brain images, aiming to isolate and define the areas corresponding to tumor tissues accurately. The segmentation process holds a pivotal role in crucial medical aspects such as diagnosis, treatment strategies, and continual monitoring of the evolution and progression of brain tumors. In recent years, significant advancements have been made in deep learning architectures, capitalizing on diverse medical imaging technologies like CT and MRI. These developments have demonstrated notable efficacy in precisely segmenting brain tumors across both 2D and 3D medical image datasets. In this regard, CNNs were the first and most commonly used in this field, and numerous methods have been proposed to automate the segmentation process. Consequently, Pereira et al. [8] introduced a CNN-based approach by investigating small kernels for deeper architecture design which had a positive effect on preventing overfitting given the low weights in the network. In addition, Zhao et al. [9] integrate a full CNN to segment different regions of a tumor by combining FCNN with a conditional random forest. Furthermore, Wang et al. [10] proposed a cascade of FCNNs to segment each subregion of the tumor by generating three binary segmentations for each

* Corresponding author.
E-mail addresses: aboussalehilyasse@gmail.com (I. Aboussaleh), jamal.riffi@gmail.com (J. Riffi), khalid.elfazazy@usmba.ac.ma (K. el Fazazy), adnan_@yahoo.com (A.M. Mahraz), hamid.tairi@usmba.ac.ma (H. Tairi).

https://doi.org/10.1016/j.rineng.2024.101892
Received 10 December 2023; Received in revised form 10 January 2024; Accepted 4 February 2024
Available online 9 February 2024
2590-1230/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

one. Besides, Aboussaleh et al. [11] exploited the features extracted from the last convolutional layer of the CNN proposed model by multiplying these features by a global average and max scalers to clarify the tumor region. Finally, a post-processing step was applied based on thresholding and the opening morphological process to eliminate insignificant pixels and improve results. Building upon this, the introduction of U-Net in biomedical image segmentation marked a significant turning point in the field. Ronneberger et al. [12]'s pioneering work in 2015 revolutionized deep learning architectures, notably impacting the way bio-medical image segmentation is approached. Whereas, several U-shaped methods have been developed achieving good results by including new techniques of machine learning and trends in deep learning architectures. Not long ago, the volumetric V-Net appeared in 2016, proposed by Milletari et al. [13] to enhance 3D medical image volumes and consider spatial relationships in all dimensions, which is particularly valuable for accurately segmenting irregularly shaped structures like tumors in volumetric medical images.

CNNs still encounter challenges in terms of long-range contextual information, hierarchical feature limitations, data inefficiency, overfitting risks, and computational complexity. In response to these challenges, hybrid architectures that combine the strengths of CNNs and Transformers are being explored to address various challenges while benefiting from the advantages of each approach. In this connection, TransUNet proposed by Chen et al. [14] was among the first architectures which introduce Vision Transformers (ViT) in the encoder part of U-Net, after extracting visual features using CNNs. Moreover, in order to segment 3D biomedical images using ViT, UNETR [15] and TransBTS [16] have been utilized to exploit the volumetric data and extract global contextual information at multiple scales in a 3D volumetric style. In this paper, we proposed a 3D hybrid encoder-decoder architecture called 3DUV-NetR+. The method takes as input 3D multimodal MR images, feeds them to 3DU-Net and V-Net models to extract semantic features at each depth from their encoders, concatenates them with their fusion to allow the hybrid model to capture both localized and contextual information, enhancing overall representation learning and apply Transformers to the obtained concatenation to obtain complex spatial relationships and dependencies across the entire image, which may not be fully captured by U-Net or V-Net alone.

This article is organized as follows: Section 2 covers the related work, presenting existing studies. In Section 3, we detail our proposed method. The experimental configuration is delineated in Section 4. Our findings are summarized, followed by a comprehensive discussion in Section 5 and a conclusion with Future Research Directions in Section 6.

2. Related works

Recently, Deep learning has proven an explosion in different computer vision tasks including medical image segmentation. Several methods based on CNNs, Attention mechanisms, and Transformers have been proposed and they succeeded in achieving a great result in this field.

2.1. Convolution neural networks methods

Convolutional Neural Networks, particularly the U-Net architecture is still used as a reference in 2D and 3D brain tumor segmentation. Therefore, Inception UDet proposed by Aboussaleh et al. [17] used the Inception module instead of the convolution block employed in UDet method [18] through the bi-directional pyramid network of the skip connection part, this modification reduced the number of parameters and time execution and increased the metrics performance of the segmented whole tumor. In addition, Rastogi et al. [19] suggest a 2D-VNet model for brain tumor segmentation and enhancing the prediction. The method showed his success by achieving high performance in terms of dice coefficient and accuracy. On the other side, contextual information aids semantic segmentation approaches to classify each pixel into predefined categories, then DL models can recognize object boundaries and precisely segment various objects according to spatial relationships between those objects. In this context, Lachinov et al. [20] proposed a deeper cascade 3D U-Net for Glioma segmentation, modifying the U-Net structure to efficiently handle multimodal MRI input. Moreover, they introduced a technique leveraging context-features obtained from the same topological model operating on scaled-down data to enhance segmentation quality. On the other hand, Casamitjana et al. [21] presented a cascade of two 3D-CNN and a later combination of their outputs to get the final segmentation, they used a modified version of V-Net consisting of convolutional blocks and residual connections reformulated according to recent findings in the literature. Additionally, Zhou et al. [22] attached a Spatial besides Channel Squeeze-and-Excitation Network (scSE-Net) to the V-Net model to calibrate the CNN image feature sampling area, which is expected to improve CNN image recognition. Also, they optimized the performance of the V-Net model in obtaining remote feature information by adding a non-local block.

2.2. Attention mechanism and transformers based methods

Motivated by the success of the attention mechanism and Transformers in numerous natural language processing tasks, they were exploited as well as possible in computer vision tasks such as classification, recognition, detection, and segmentation. In this regard, Zhang et al. [23] proposed an Attention Gate ResU-Net for automatic MRI brain tumor segmentation. They employed a residual block and an attention gate with a single U-Net architecture added into the skip connection part, moreover, Aboussaleh et al. [24] introduced a decoder based on the attention mechanism after fusing the extracted features of three different pre-trained models Vgg19, ResNets50, and MobileNetV2 to finally segment the different sub-regions of a brain tumor. On the other option, Vision Transformers showed a major impact on medical image segmentation and multiple studies have been enhanced, some incorporated Transformers with CNN-based U-Net, and others used mainly standalone Transformers for creating backbone U-Net architectures. Xie et al. [25] bridged CNN and Transformers to propose a new framework for medical image segmentation. The CNN is constructed to extract feature representation and the proposed deformable Transformers are used for long-range dependency on these features, including the deformable self-attention mechanism. TransFuse proposed by Zhang et al. [26] aims to combine CNN and Transformers to extract both low-level spatial features and high-level semantic features and fuse them in a parallel way using their novel fusion technique BiFusion. Further, Wang et al. [27] replaced the original skip connection of U-Net with a modified one CTrans module, including a channel-wise cross-fusion transformer and channel-wise cross attention for aggregating multi-scale features adaptively and forward the merged multi-scale features to the decoder channel-by-channel. Not long ago, the Swin Transformer (ST) proposed by Liu et al. [28] showed a significant improvement in segmentation results by applying their appropriate hierarchical transformer which has a high features representation of object concepts, computed by shifted windows. Based on this idea, Hatamizadeh et al. [29] exploited a block of ST in the encoder part of his swin UNETR method, then, they fed the encoded features extracted to a CNN-decoder via skip connection at each resolution from five employed. A volumetric medical image segmentation has been introduced by Zhou et al. [30] via 3D Transformers, they realized a new combination technique with convolution and self-attention operations, used both local and global features representation and proposed to skip attention instead of the classical skip connection that often relies on the sum or merge operations. Wei et al. [31] proposed a high-resolution swin transformer by creating a deep architecture containing several stages, except the last one, each stage generates multi-resolution feature maps, by using n block of transformers in the corresponding n stage. All the stages are sequentially connected and a multi-resolution fusion block will apply to each stage output, and the outputs will be concatenated after up-sampling low-resolution features to get the final segmentation after passing the concatenated

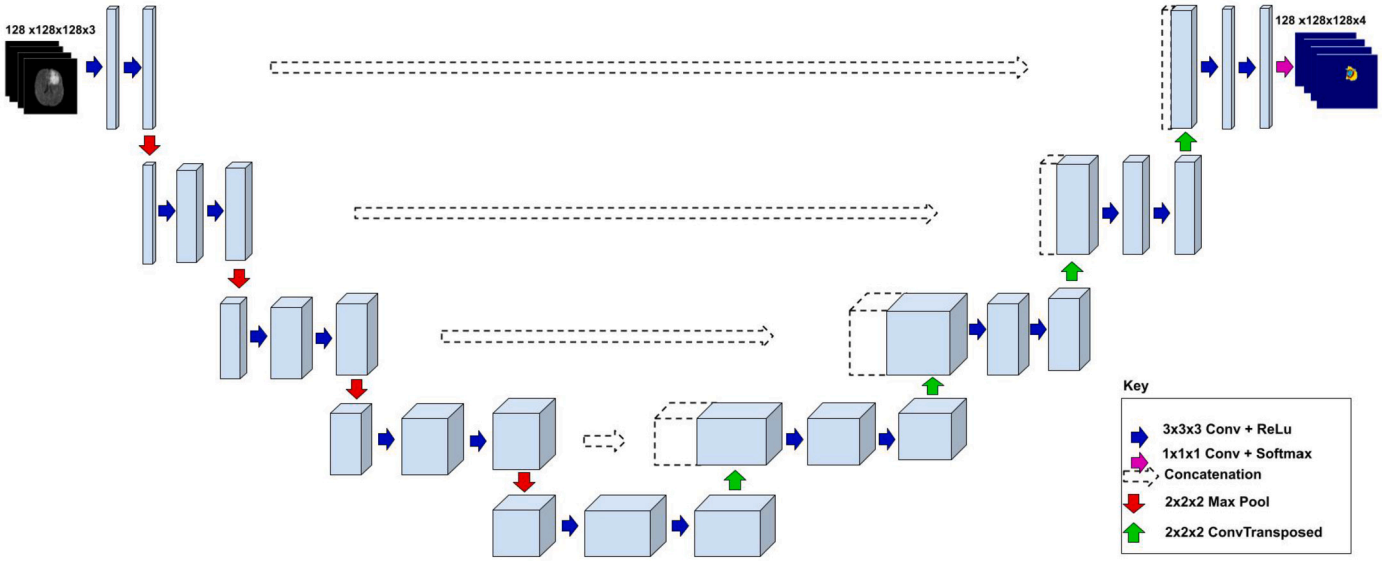


Fig. 1. The structure of the 3D U-Net architecture used in our proposed method.

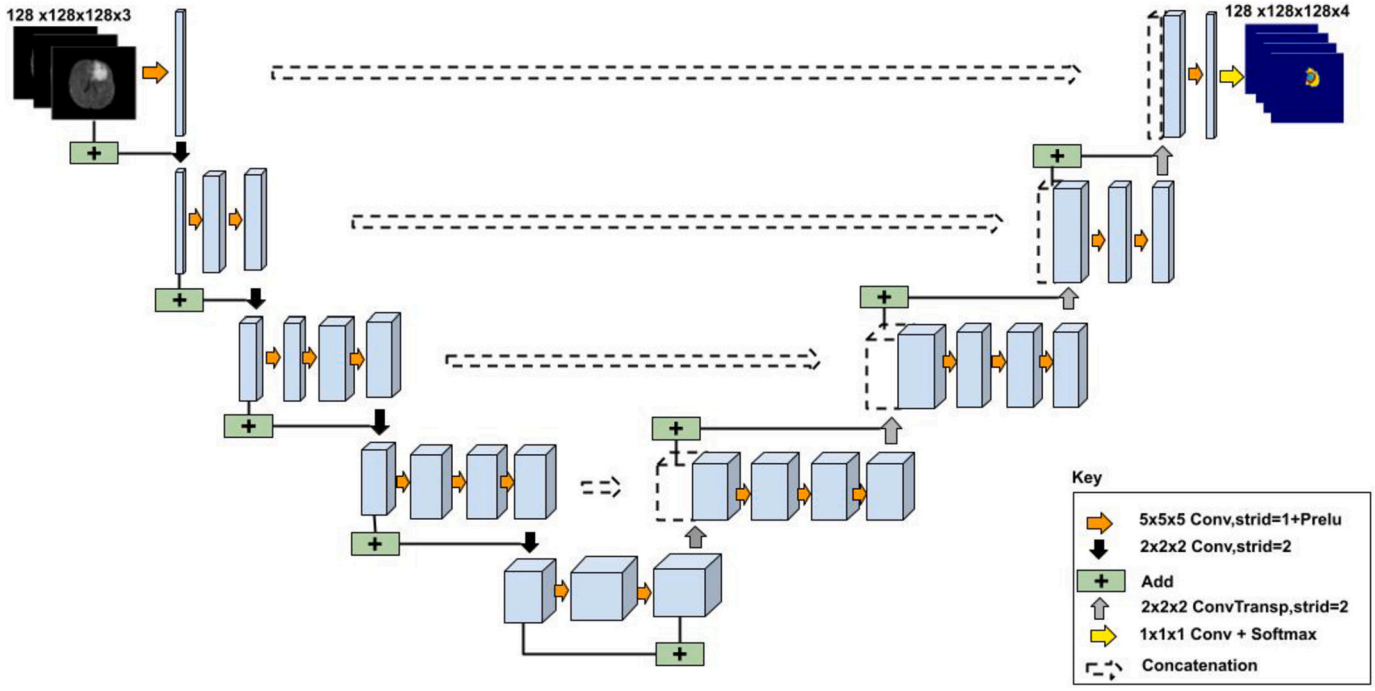


Fig. 2. The structure of the V-Net architecture used in our proposed method.

feature through a series of residual blocks, a patch expanding block, and a convolution block.

3. Proposed method

In this section, we explain in detail our proposed method 3DUV-NetR+ illustrated in Fig. 3, including both encoder and decoder parts. The encoder path is created from the encoders of 3DU-Net and V-Net based on the feature extracted in each depth, while the decoder path uses feature fusion and Transformers for more meaningful features.

3.1. Encoder

The proposed encoder of 3DUV-NetR+ is derived from the following 3DU-Net and V-Net:

- **3DU-Net**: The 3D U-Net architecture, Fig. 1, extends the principles of U-Net into three dimensions, utilizing convolutional layers for volumetric data. It encompasses an encoder-decoder structure with skip connections designed specifically for 3D semantic segmentation tasks. The encoder, akin to U-Net, consists of blocks involving repeated 3D convolutional operations followed by max-pooling to downsample the input volume. This process incrementally increases the number of filters to capture spatial information. The decoder part of 3D U-Net employs transposed convolutions for upsampling, gradually reducing the number of filters while reconstructing the segmentation map. Similarly, skip connections are employed to establish links between corresponding encoder and decoder layers, facilitating the retention and restoration of spatial features essential for accurate 3D segmentation. The activation functions primarily employ ReLU, except for the last layer,

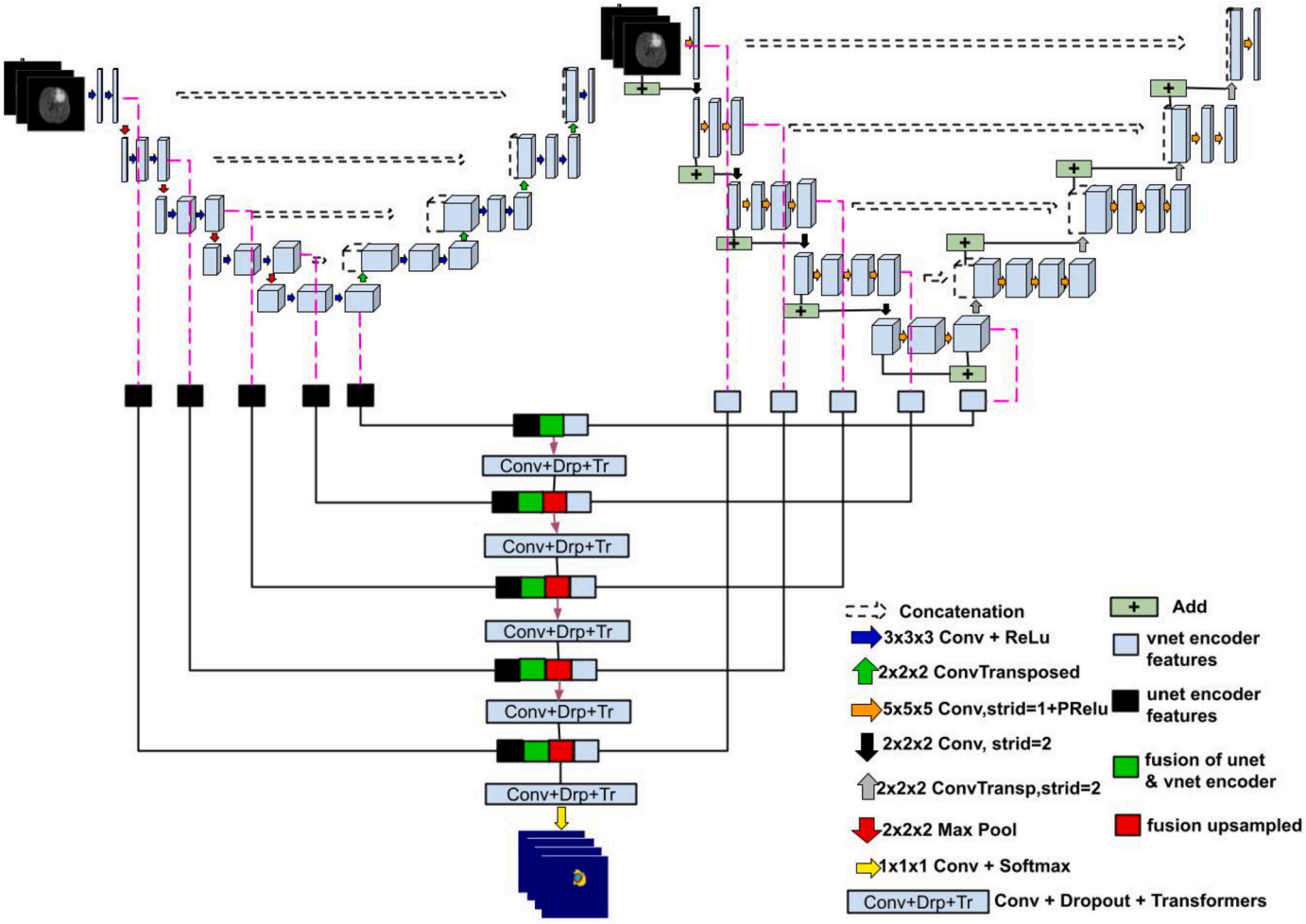


Fig. 3. Overview of the proposed 3DUV-NetR+ architecture.

which typically uses a sigmoid activation to generate the final segmentation map. The primary architectural layout of the proposed 3DU-Net is delineated in the Table 1.

- **V-Net**: is an extension of the U-Net architecture, designed to handle 3D medical image segmentation tasks. It incorporates the same U-shaped architecture as UNet, but it's adapted for 3D volumes. V-Net, Fig. 2, introduces several modifications to the UNet architecture to work with 3D data, including 3D convolutional layers and 3D pooling operations besides using convolutional residual units to capture features more effectively across multiple spatial scales. These modifications allow the network to capture spatial dependencies in all three dimensions, making it more suitable for volumetric segmentation tasks.
  Similar to U-Net, V-Net introduces skip blocks in the encoding pathway. These blocks involve skipping connections directly from one encoding layer to another, allowing for improved information flow between different resolution levels and enhancing feature learning across multiple scales. The main architecture of the proposed V-Net is shown in Table 2.

3.2. Decoder

After extracting features from the encoder part, the decoder part of 3DUV-NetR+ consists of three main contributions feature fusion, enhanced representation learning, and Transformers at each upsampling level, respectively. Then, a final 3D convolution block of 4 filters referring to different tumor sub-regions with 1x1x1 size is applied to generate the final segmented tumor.

Feature fusion: These extracted features from both 3DU-Net and V-Net encoders are concatenated or fused to create a combined representation. This fusion aims to leverage the strengths of both models, enabling the hybrid model to benefit from both localized and contextual information.

Enhanced representation learning: By combining the semantic features and their fusion, the hybrid model aims to enhance overall representation learning, allowing it to capture a broader range of information and feature representations.

Transformers: Transformers are applied to the combined feature representations obtained from the fusion process. The goal is to leverage the capabilities of transformers in capturing complex spatial relationships and dependencies across the entire 3D image volume, which might not be fully captured by either 3DU-Net or V-Net alone. Then, Transformers use an attention mechanism to process input data, allowing them to weigh and combine different parts of the input, making them excellent at understanding context. We used a stack of L identical layers, used in the encoder Transformer, shown in Fig. 4, that consists of two main sub-layers; Multi-Head Self Attention (MSA) and Multi-Layer Perceptron (MLP), a Layer Normalization (LN) is applied before each sub-layer of MSA and MLP, and a residual connection is applied after each one. The output Z_l of the l-layer is computed as:

$$\hat{Z}_l = MSA(LN(Z_{l-1})) + Z_{l-1} \quad (1)$$

$$Z_l = MLP(LN(\hat{Z}_l)) + \hat{Z}_l \quad (2)$$

where $l \in \{1, 2, ..., L\}$ and $\hat{Z}_l$ the output of the MSA sub-layer.

Table 1
Proposed 3D U-Net architecture with 5 depths in encoder and decoder.

Layer	Input Shape	Output Shape
Input	128x128x128x3	-
Conv3D	128x128x128x3	128x128x128x8
Conv3D	128x128x128x8	128x128x128x8
MaxPooling3D	64x64x64x8	64x64x64x8
Conv3D	64x64x64x8	64x64x64x16
Conv3D	64x64x64x16	64x64x64x16
MaxPooling3D	32x32x32x16	32x32x32x16
Conv3D	32x32x32x16	32x32x32x32
Conv3D	32x32x32x32	32x32x32x32
MaxPooling3D	16x16x16x32	16x16x16x32
Conv3D	16x16x16x32	16x16x16x64
Conv3D	16x16x16x64	16x16x16x64
MaxPooling3D	8x8x8x64	8x8x8x64
Conv3D	8x8x8x64	8x8x8x128
Conv3D	8x8x8x128	8x8x8x128
UpSampling3D	8x8x8x128	16x16x16x128
Concatenate	16x16x16x128	16x16x16x192
Conv3D	16x16x16x192	16x16x16x64
Conv3D	16x16x16x64	16x16x16x64
UpSampling3D	16x16x16x64	32x32x32x64
Concatenate	32x32x32x64	32x32x32x96
Conv3D	32x32x32x96	32x32x32x32
Conv3D	32x32x32x32	32x32x32x32
UpSampling3D	32x32x32x32	64x64x64x32
Concatenate	64x64x64x32	64x64x64x48
Conv3D	64x64x64x48	64x64x64x16
Conv3D	64x64x64x16	64x64x64x16
UpSampling3D	64x64x64x16	128x128x128x16
Concatenate	128x128x128x16	128x128x128x24
Conv3D	128x128x128x24	128x128x128x8
Conv3D	128x128x128x8	128x128x128x8
Output	128x128x128x8	128x128x128x4

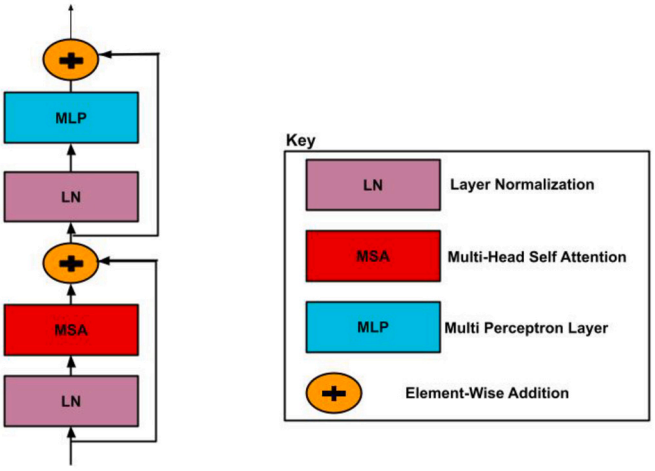


Fig. 4. Transformers block using in the decoder part of our proposed 3DUV-NetR+ method.

4. Experiment

4.1. Data

BraTS 2020 [32–34] consists of 369 training scans and 128 scans for validation. Each sample (scan) is composed of four 3D MRI modalities: a) native (T1) b) post-contrast T1-weighted (T1Gd), c) T2-weighted (T2), and d) T2 Fluid-attenuated Inversion Recovery (T2-FLAIR). These modalities are rigidly aligned, resampled to a $1 \times 1 \times 1mm$ isotropic resolution, and skull-stripped. $240 \times 240 \times 155$ is the size of the input image. The information was gathered from numerous institutions utilizing a variety of MRI scanners. The imaging datasets underwent manual segmentation by one to four raters, adhering to a uniform annotation protocol. These annotations, inclusive of the GD-enhancing tumor (ET — label 4), peritumoral edema (ED — label 2), and the necrotic and non-enhancing tumor core (NCR/NET — label 1) were meticulously approved by experienced neuro-radiologists.

Table 2
Proposed modified V-Net architecture.

Layer	Input Size	Output Size	S/F
Input	128x128x128x3	-	-
Conv3D	128x128x128x3	128x128x128x8	1/5x5x5
Conv3D	128x128x128x8	128x128x128x8	1/5x5x5
Conv3D	128x128x128x8	64x64x64x8	2/2x2x2
Conv3D	64x64x64x8	64x64x64x16	1/5x5x5
Conv3D	64x64x64x16	32x32x32x16	2/2x2x2
Conv3D	32x32x32x16	32x32x32x32	1/5x5x5
Conv3D	32x32x32x32	16x16x16x32	2/2x2x2
Conv3D	16x16x16x32	16x16x16x64	1/5x5x5
Conv3D	16x16x16x64	8x8x8x64	2/2x2x2
Conv3D	8x8x8x64	8x8x8x128	1/5x5x5
Deconv3D	8x8x8x128	16x16x16x128	2/2x2x2
Concatenate	16x16x16x128	16x16x16x192	-
Conv3D	16x16x16x192	16x16x16x64	1/5x5x5
Conv3D	16x16x16x64	16x16x16x64	1/5x5x5
Deconv3D	16x16x16x64	32x32x32x64	2/2x2x2
Concatenate	32x32x32x64	32x32x32x96	-
Conv3D	32x32x32x96	32x32x32x32	1/5x5x5
Conv3D	32x32x32x32	32x32x32x32	1/5x5x5
Deconv3D	32x32x32x32	64x64x64x32	2/2x2x2
Concatenate	64x64x64x32	64x64x64x48	-
Conv3D	64x64x64x48	64x64x64x16	1/5x5x5
Conv3D	64x64x64x16	64x64x64x16	1/5x5x5
Deconv3D	64x64x64x16	128x128x128x16	2/2x2x2
Concatenate	128x128x128x16	128x128x128x24	-
Conv3D	128x128x128x24	128x128x128x8	1/5x5x5
Conv3D	128x128x128x8	128x128x128x8	1/5x5x5
Output	128x128x128x8	128x128x128x4	-

4.2. Data preparation and preprocessing

We transformed each patient's size of the BraTS 2020 dataset from 240×240×155 to 128×128×128 for 3D segmentation. We eliminated the first 15 slices and the last 12 ones to keep the slices when the tumor appeared well. Each slice of different modalities was cropped into 128×128×128 to eliminate some non-significant background pixels. In the pre-processing step, we normalized the obtained 3D image by subtracting its mean and dividing it by its standard deviation.

4.3. Evaluation metrics

Several evaluation metrics concern the semantic segmentation task, to demonstrate the performance of our model, we used different performance indicators: Accuracy (Acc), Dice Similarity coefficient (DSC), Intersection over Union (IoU), and Hausdorff Distance:

- **DSC**: The DSC represents the overlapping of predicted segmentation with the manually segmented output label and is computed as:

$$DSC = \frac{2 \times |G \cap S|}{|G| + |S|} \tag{3}$$

- **HD95**: The Hausdorff95 distance measures the distance between the surface of the real area and the predicted area which is more sensitive to the segmented boundary defined as:

$$d_H(G,S) = \max\left\{ \sup_{g \in G} \inf_{s \in S} d(g,s), \sup_{s \in S} \inf_{g \in G} d(g,s) \right\}, \tag{4}$$

where G and S are predicted segmentation and the manually segmented output labe respectively and d is the distance metric on the space.

4.4. Implementation details

In this experiment, TensorFlow has been used to build our models, including basic models, meta-models, and other relative models introduced in the comparison results section, we trained all the models in an NVIDIA QUADRO P6000 GPU with 24 GH of RAM. We used three MRI modalities T1ce, T2, and Flair as the most recommended ones for brain tumor segmentation tasks due to their Value in detecting and identifying different sub-regions of the brain tumor. Additionally, the Basic models and the meta-model are trained from scratch for 200 epochs with a batch size equal to 2 in both the training and test phases. For the loss function used, we took the combination of categorical Focal Loss and Dice Loss functions to get a faster convergence and better performance, optimized by Adam optimizer with weight decay $1e-4$ and a learning rate equal to 0.001.

$$\text{Loss} = \text{Categorical Focal Loss} + \text{Dice Loss}$$

$$\text{Loss} = -\sum_{i=1}^{N} \left(g_i \cdot \left(1-p_i\right)^{\gamma} \cdot \log(p_i)\right) + 1 - \frac{2 \cdot \sum_i (p_i \cdot g_i)}{\sum_i p_i^2 + \sum_i g_i^2} \tag{5}$$

Where g_i, p_i, and γ represent ground truth, predicted probability, and focal loss parameter, respectively.
The focal loss parameter γ controls the degree of focus on hard-to-classify examples and its default value is 2.0.

5. Results and discussion

In this section, we showcase the results achieved by our 3DUV-NetR+ approach concerning the recommended metrics. We compared these results with other state-of-the-art methods. Additionally, we provide visual representations of qualitative results to assess our model's performance. Furthermore, an ablation study is conducted to scrutinize the individual techniques and contributions integrated into 3DUV-NetR+.

5.1. Ablation study

Table 3 demonstrates performance improvements in terms of higher DSC scores and lower HD95 values as methods progress from baseline approaches (3DU-Net and V-Net) to more enhanced architectures (3DUV-Net, 3DUV-Net+, 3DUV-NetR+). The hybrid approach 3DUV-Net that merges encoder parts of 3DU-Net and V-Net represents competitive performance, especially for the WT sub-region by achieving 0.898 and 5.2 mm in terms of DSC and HD95mm, respectively. Furthermore, 3DUV-Net+ outperforms the key approaches, while, the 3DUV-NetR+ method exhibits the highest DSC scores by achieving 0.919, 0.828, and 0.817 for WT, TC, and ET respectively, and manifests the lowest HD95 values of 4.9 mm, 6.0 mm, and 3.8 mm for WT, TC, and ET respectively. The inclusion of Transformer blocks within each decoder level in 3DUV-NetR+ appears to significantly contribute to its improved segmentation performance, enabling better feature representation and capturing complex spatial relationships within the images.
Associated with this study, the graphical depiction in Fig. 6 highlights the consistently positive performance of all models, showcasing notable progress throughout their development. This emphasizes the significant impact of our contributions on enhancing the overall results. Furthermore, achieving highly accurate results on the BraTS 2020 dataset serves as a significant step towards validating the practical application of our methodology in real clinical scenarios and helps bridge the gap between algorithmic performance and clinical relevance.

Table 3
Ablation study of 3DUV-NetR+ in terms of DSC and HD95 on the test sub-set of BraTS 2020 training dataset.

	DSC%			HD95*mm*		
Method	WT	TC	ET	WT	TC	ET
3DU-Net	0.896	0.807	0.766	6.6	6.4	5.1
V-Net	0.888	0.787	0.798	5.3	6.7	4.4
3DUV-Net	0.898	0.805	0.795	5.2	6.6	4.8
3DUV-Net+	0.906	0.810	0.800	5.0	6.3	4.2
3DUV-NetR+	0.919	0.828	0.817	4.9	6.0	3.8

Table 4
Comparison study of our proposed methods 3DUV-NetR+ with other SOTA methods on BraTS datasets.

	DSC%			HD95*mm*		
Method	WT	TC	ET	WT	TC	ET
Pereira et al [8]	88.00	83.00	77.00	-	-	-
Aboussaleh et al [11]	82.35	-	-	-	-	-
Lachinov et al [20]	90.00	83.00	77.00	-	-	-
Casamitjana et al [21]	87.00	71.00	64.00	-	-	-
scSE-NL V-Net [22]	82.00	76.00	65.00	-	-	-
CU-Net [35]	88.80	83.10	76.80	-	-	-
Inception-UDet [17]	87.90	-	-	-	-	-
Zhang et al [23]	88.80	78.10	75.00	6.8	9.3	3.8
Aboussaleh et al [24]	87.41	80.69	70.33	-	-	-
Wang et al [36]	90.50	83.80	78.60	3.9	6.5	3.3
TransUNet [14]	70.60	68.40	54.20	14.0	14.5	10.0
TransBTS [16]	77.90	73.50	57.40	10.0	8.9	9.9
UNETR [15]	78.90	76.50	58.10	8.2	8.8	9.3
nnFormer [30]	91.30	86.00	81.80	3.8	4.5	3.9
3DUV-NetR+(ours)	91.95	82.80	81.70	4.9	6.0	3.8

5.2. Comparison study with the SOTA methods

Table 4 provides a comprehensive overview of the comparative performance of different segmentation methods on BraTS datasets, emphasizing the superior performance of the proposed 3DUV-NetR+ method. The comparison study covers CNN-based methodologies [8,11] U-Net-like architectures [20–22,35,17], Attention mechanism-based approaches [23,24,10], and Transformer-based methodologies [14,16,15, 30]. The proposed method accomplished the highest DSC of WT, besides a competitive performance of nnFormer and Wang et al. approach, which are based on the self-attention mechanism. They also fulfilled the best score for TC by achieving 86% and 83.80%, respectively. We reached the second top performance of 81.70% for ET, which is very close to the top one at 81.80%. On the other hand, the HD95 values achieved by 3DUV-NetR+ are notably lower (indicating better boundary prediction) compared to several methods in the table. However, certain methods, such as Wang et al. and nnFormer, exhibit even lower HD95 values, indicating slightly better performance in accurately delineating tumor boundaries compared to 3DUV-NetR+. Fig. 7 illustrates the graphical representation of the results presented in Table 4 using different versions of the BraTS dataset. The findings denote that the proposed outperforms the SOTA methods mentioned for the WT sub-region and competitive results for other sub-regions. However, it's essential to recognize several limitations associated with our approach, particularly, the dataset employed is diversity-limited in Tumor Types and Characteristics, it primarily focuses on gliomas which may not fully encompass other rare or less common types of brain tumors, moreover, the distribution of tumor classes within the dataset might not

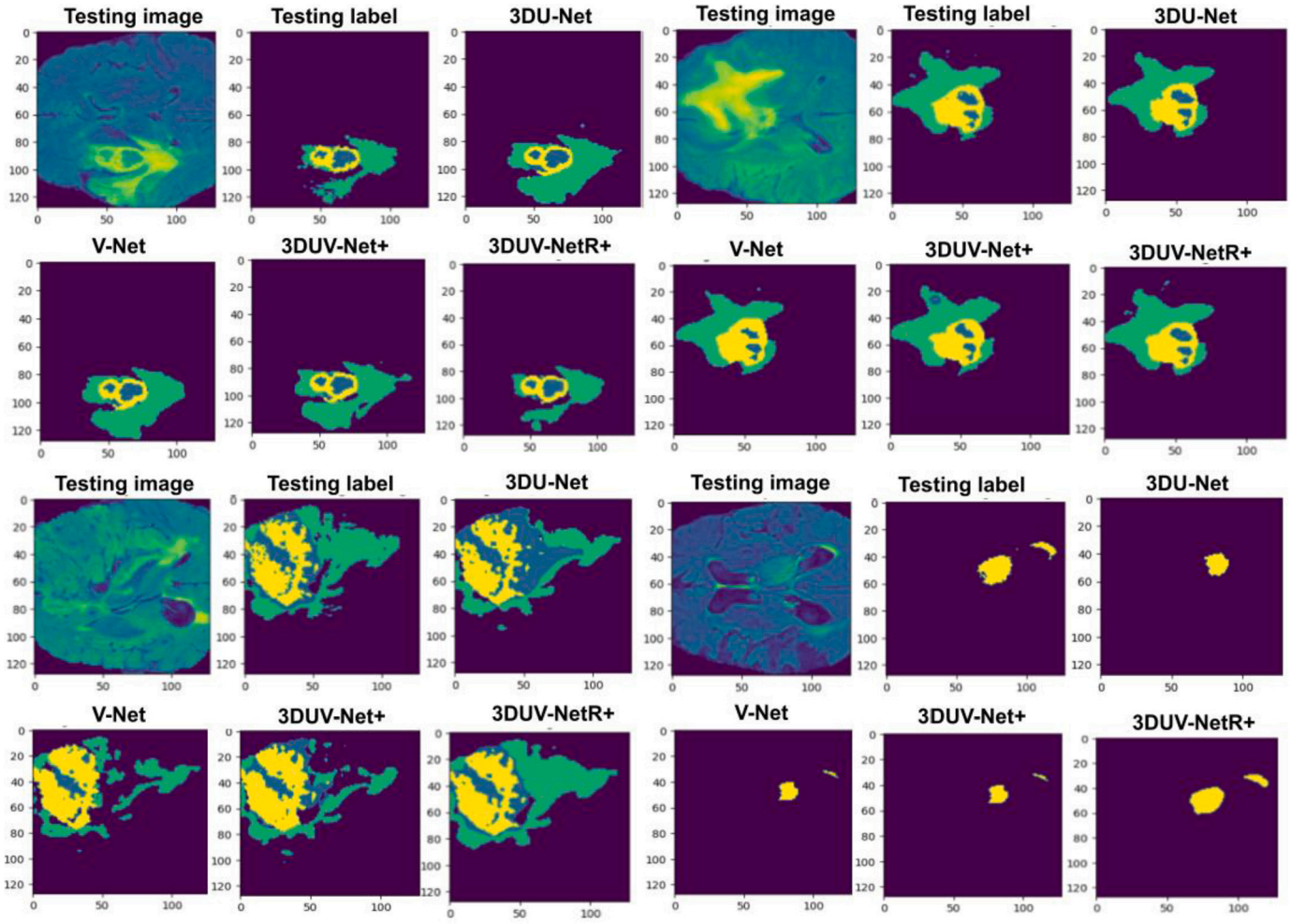


Fig. 5. Visualization segmentation results of some samples of BraTS 2020 training dataset. From left to right: Ground Truth, 3DU-Net, V-Net, 3DUV-Net+, and 3DUV-NetR+ results overlaid on the Flair image. Each color represents a tumor region: green—whole tumor, yellow—core tumor, and bleu—enhancing tumor.

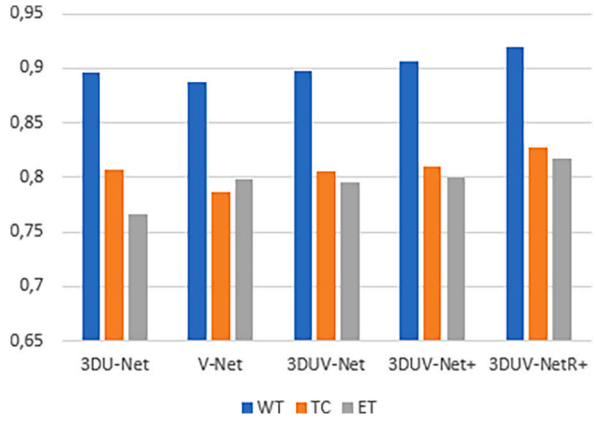


Fig. 6. Graphical representation of ablation comparison in terms of DSC on BraTS 2020 dataset.

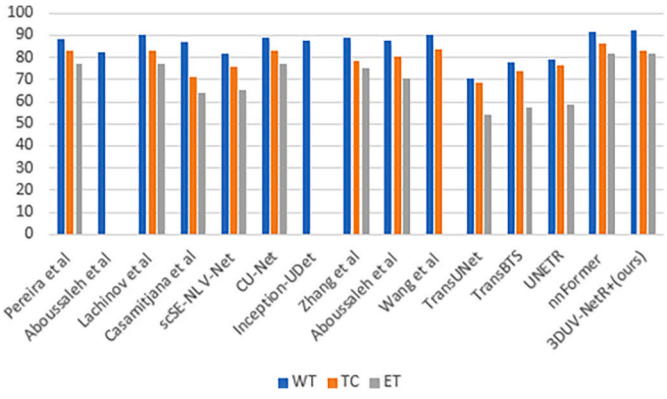


Fig. 7. Graphical representation that shows DSC comparison of the proposed model and the SOTA deep segmentation models on different BraTS datasets.

be uniform. This imbalance could lead to biases, causing the model to be more proficient in segments of the data that are well-represented while performing poorly in underrepresented or minority classes. In addition, despite efforts to ensure consistency through multiple raters and neuro-radiologist approvals, inherent variability in manual annotations might exist and the absence of clinical information or contextual data associated with each case, such as patient history, treatment plans, or outcomes limits the model's ability to learn from and incorporate clinical context, potentially affecting the practical applicability of the model's outputs in real clinical scenarios. Moreover, the complexity of the segmentation task for magnetic resonance images is heightened by the absence of powerful computing resources, consequently increasing the time required for completion.

Moreover, a comparison study of our proposed model and his sub-models has been represented in Table 5 in terms of training time and number of parameters, then the highest obtained results reflect one of our limitations in terms of computational performance in the training

Table 5
Ablation study of computational performance in terms of training time and number of parameters for each model.

Method	Training-time	Number of Parameters (M)
3D-UNet	3.05 h	5.6
VNet	4.2 h	11
3DUV-Net	4.5 h	11.6
3DU-VNet+	4.5 h	11.6
3DUV-NetR+	4.6 h	11.7

phase. Optimization is required to further improve the outcomes and minimize the execution time.

5.3. Visualization of results

Visualization results are illustrated in Fig. 5 employing three modalities (Flair, T1ce, T2) as input to generate the final 3D mask with different sub-regions of tumor discriminated by a precis color for each tumor part. We can see that our method shows good performance compared to the basic models, they almost reach the true label which emphasizes the impact of the fusion of features extracted from the encoder of basic models 3DUNet and V-Net in addition to employing Transformers decoder, to achieve better performance and visualization.

6. Conclusion and perspective

This study substantiates a strong correlation between advancements in scientific research within the medical realm and the practice of medical data analysis. Specifically, brain tumor segmentation stands as an intricate domain demanding meticulous handling and precision throughout Therefore, in this manuscript, we meticulously tried to improve the Brain tumor segmentation architectures by introducing 3DUV-NetR+ architecture that integrates the strengths of both 3DU-Net and V-Net models by extracting features from their encoders, fusing these features, and subsequently applying transformers to enhance the model's ability to capture complex spatial relationships and dependencies for more accurate and detailed segmentation in 3D multimodal MR images. We conducted training and evaluation of our model on the BraTS2020 dataset, using ground truths for comparison with the state-of-the-art models mentioned previously. The experimental results strongly indicate the effectiveness and high performance of our architecture in segmentation tasks. Moving forward, our future work will primarily concentrate on enhancing the previous results, specifically tumor core and enhancing edema sub-regions, by creating deeper models based on the new architectures and techniques of deep learning. Additionally, we plan to create our online platform 3D Segmentation System Assistant Radiologists (3D2SAR) to help medical specialists in data annotation and augment medical labeled data that will be used to improve segmentation models in the medical image analysis field.

CRediT authorship contribution statement

Ilyasse Aboussaleh: Writing – original draft, Methodology, Conceptualization. **Jamal Riffi:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Khalid el Fazazy:** Writing – review & editing. **Adnane Mohamed Mahraz:** Writing – review & editing. **Hamid Tairi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is available on Kaggle platform.

References

[1] S. Sreelakshmi, G. Malu, E. Sherly, R. Mathew, M-net: an encoder-decoder architecture for medical image analysis using ensemble learning, Results Eng. 17 (2023) 100927.
[2] S. Arvind, J.V. Tembhurne, T. Diwan, P. Sahare, Improvised light weight deep cnn based u-net for the semantic segmentation of lungs from chest x-rays, Results Eng. 17 (2023) 100929.
[3] J. Wang, R. Zietal, A. Arase, G. Couzens, P. Pivonka, D. Fontanarosa, Computer-assisted pre-operative automatic segmentation and registration tool for malunited radius osteotomy: a proof-of-concept study, Results Eng. 19 (2023) 101295.
[4] M. Oulhadj, J. Riffi, C. Khodriss, A.M. Mahraz, A. Bennis, A. Yahyaouy, F. Chraibi, M. Abdellaoui, I.B. Andaloussi, H. Tairi, Diabetic retinopathy prediction based on wavelet decomposition and modified capsule network, J. Digit. Imag. (2023) 1–13.
[5] M. Hussain, D. Koundal, J. Manhas, Deep learning-based diagnosis of disc degenerative diseases using mri: a comprehensive review, Comput. Electr. Eng. 105 (2023) 108524.
[6] R.G. Poola, L. Pl, et al., Covid-19 diagnosis: a comprehensive review of pre-trained deep learning models based on feature extraction algorithm, Results Eng. (2023) 101020.
[7] X. Luo, B. Du, P. Gui, D. Zhang, W. Hu, A hunger games search algorithm with opposition-based learning for solving multimodal medical image registration, Neurocomputing 540 (2023) 126204.
[8] S. Pereira, A. Pinto, V. Alves, C.A. Silva, Brain tumor segmentation using convolutional neural networks in mri images, IEEE Trans. Med. Imaging 35 (5) (2016) 1240–1251.
[9] X. Zhao, Y. Wu, G. Song, Z. Li, Y. Fan, Y. Zhang, Brain tumor segmentation using a fully convolutional neural network with conditional random fields, in: Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: Second International Workshop, BrainLes 2016, with the Challenges on BRATS, ISLES and mTOP 2016, Held in Conjunction with MICCAI 2016, Athens, Greece, October 17, 2016, Revised Selected Papers 2, Springer, 2016, pp. 75–87.
[10] G. Wang, W. Li, S. Ourselin, T. Vercauteren, Automatic brain tumor segmentation using cascaded anisotropic convolutional neural networks, in: Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: Third International Workshop, BrainLes 2017, Held in Conjunction with MICCAI 2017, Quebec City, QC, Canada, September 14, 2017, Revised Selected Papers 3, Springer, 2018, pp. 178–190.
[11] I. Aboussaleh, J. Riffi, A.M. Mahraz, H. Tairi, Brain tumor segmentation based on deep learning's feature representation, J. Imaging 7 (12) (2021) 269.
[12] O. Ronneberger, P. Fischer, T. Brox, U-net: convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Proceedings, Part III 18, Munich, Germany, October 5-9, 2015, Springer, 2015, pp. 234–241.
[13] F. Milletari, N. Navab, S.-A. Ahmadi, V-net: fully convolutional neural networks for volumetric medical image segmentation, in: 2016 Fourth International Conference on 3D Vision (3DV), IEEE, 2016, pp. 565–571.
[14] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A.L. Yuille, Y. Zhou, Transunet: transformers make strong encoders for medical image segmentation, arXiv preprint, arXiv:2102.04306, 2021.
[15] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H.R. Roth, D. Xu, Unetr: transformers for 3d medical image segmentation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 574–584.
[16] W. Wang, C. Chen, M. Ding, H. Yu, S. Zha, J. Li, Transbts: multimodal brain tumor segmentation using transformer, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Proceedings, Part I 24, Strasbourg, France, September 27–October 1, 2021, Springer, 2021, pp. 109–119.
[17] I. Aboussaleh, J. Riffi, A.M. Mahraz, H. Tairi, Inception-udet: an improved u-net architecture for brain tumor segmentation, Ann. Data Sci. (2023) 1–23.
[18] N.V. Keetha, C.S.R. Annavarapu, et al., U-det: a modified u-net architecture with bidirectional feature network for lung nodule segmentation, arXiv preprint, arXiv: 2003.09293, 2020.
[19] D. Rastogi, P. Johri, V. Tiwari, Brain tumor segmentation and tumor prediction using 2d-vnet deep learning architecture, in: 2021 10th International Conference on System Modeling & Advancement in Research Trends (SMART), IEEE, 2021, pp. 723–732.
[20] D. Lachinov, E. Vasiliev, V. Turlapov, Glioma segmentation with cascaded unet, in: International MICCAI Brainlesion Workshop, Springer, 2018, pp. 189–198.
[21] A. Casamitjana, M. Catà, I. Sánchez, M. Combalia, V. Vilaplana, Cascaded v-net using roi masks for brain tumor segmentation, in: International MICCAI Brainlesion Workshop, Springer, 2017, pp. 381–391.
[22] J. Zhou, J. Ye, Y. Liang, J. Zhao, Y. Wu, S. Luo, X. Lai, J. Wang, scse-nl v-net: a brain tumor automatic segmentation method based on spatial and channel "squeeze-and-excitation" network with non-local block, Front. Neurosci. 16 (2022) 916818.
[23] J. Zhang, Z. Jiang, J. Dong, Y. Hou, B. Liu, Attention gate resu-net for automatic mri brain tumor segmentation, IEEE Access 8 (2020) 58533–58545.
[24] I. Aboussaleh, J. Riffi, K.E. Fazazy, M.A. Mahraz, H. Tairi, Efficient u-net architecture with multiple encoders and attention mechanism decoders for brain tumor segmentation, Diagnostics 13 (5) (2023) 872.

[25] Y. Xie, J. Zhang, C. Shen, Y. Xia Cotr, Efficiently bridging cnn and transformer for 3d medical image segmentation, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Proceedings, Part III 24, Strasbourg, France, September 27–October 1, 2021, Springer, 2021, pp. 171–180.
[26] Y. Zhang, H. Liu, Q. Hu Transfuse, Fusing transformers and cnns for medical image segmentation, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Proceedings, Part I 24, Strasbourg, France, September 27–October 1, 2021, Springer, 2021, pp. 14–24.
[27] H. Wang, P. Cao, J. Wang, O.R. Zaiane, Uctransnet: rethinking the skip connections in u-net from a channel-wise perspective with transformer, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 36, 2022, pp. 2441–2449.
[28] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.
[29] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H.R. Roth, D. Xu, Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images, in: International MICCAI Brainlesion Workshop, Springer, 2021, pp. 272–284.
[30] H.-Y. Zhou, J. Guo, Y. Zhang, L. Yu, L. Wang, Y. Yu, nnformer: interleaved transformer for volumetric segmentation, arXiv preprint, arXiv:2109.03201, 2021.
[31] C. Wei, S. Ren, K. Guo, H. Hu, J. Liang, High-resolution swin transformer for automatic medical image segmentation, Sensors 23 (7) (2023) 3420.
[32] B.H. Menze, A. Jakab, S. Bauer, J. Kalpathy-Cramer, K. Farahani, J. Kirby, Y. Burren, N. Porz, J. Slotboom, R. Wiest, et al., The multimodal brain tumor image segmentation benchmark (brats), IEEE Trans. Med. Imaging 34 (10) (2014) 1993–2024.
[33] S. Bakas, H. Akbari, A. Sotiras, M. Bilello, M. Rozycki, J.S. Kirby, J.B. Freymann, K. Farahani, C. Davatzikos, Advancing the cancer genome atlas glioma mri collections with expert segmentation labels and radiomic features, Sci. Data 4 (1) (2017) 1–13.
[34] S. Bakas, M. Reyes, A. Jakab, S. Bauer, M. Rempfler, A. Crimi, R.T. Shinohara, C. Berger, S.M. Ha, M. Rozycki, et al., Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the brats challenge, arXiv preprint, arXiv:1811.02629, 2018.
[35] H. Liu, X. Shen, F. Shang, F. Ge, F. Wang, Cu-net: cascaded u-net with loss weighted sampling for brain tumor segmentation, in: Multimodal Brain Image Analysis and Mathematical Foundations of Computational Anatomy: 4th International Workshop, MBIA 2019, and 7th International Workshop, MFCA 2019, Held in Conjunction with MICCAI 2019, Proceedings 4, Shenzhen, China, October 17, 2019, Springer, 2019, pp. 102–111.
[36] G. Wang, W. Li, S. Ourselin, T. Vercauteren, Automatic brain tumor segmentation based on cascaded convolutional neural networks with uncertainty estimation, Front. Comput. Neurosci. 13 (2019) 56.